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22EC301
PROBABILITY, STATISTICS AND RANDOM PROCESS

UNIT IV

DESIGN OF EXPERIMENTS AND CONTROL CHART

CONTROL CHARTS FOR ATTRIBUTES

(p, and np charts).

Control chart

A control chart provides a basis for deciding whether the variance in the output due to random causes or due to assignable causes. It will assist us in making decisions whether to adjust the process or not.

A control chart is designed to display successive measurements of a process with a Centre line and control limits.

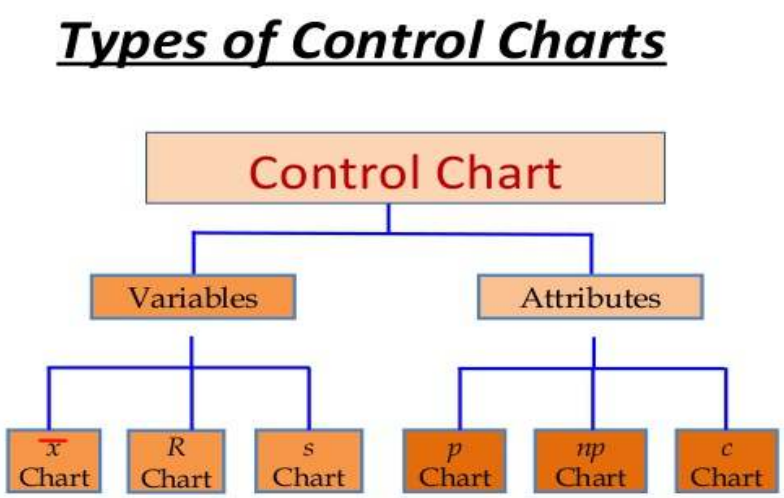
The control limits are above and below the Centre line and are equidistant from the Centre line and are known as Upper control limit (UCL) and Lower control limit (LCL).

The control charts help us decides whether the process of production is in control or not.

Two types of control charts

(i) Control charts of variables

(ii) Control charts for attributes



Control charts for attributes

To control the quality of certain product whose attributes are available, the following control chats are used

(i) np- chart of number of defective

(ii) p-chart for proportion of defective

(iii) c- Chart for the number of defects in a unit.

Np-chart (or) control chart for number of defectives:

np CHART FORMULA

The centre line and upper and lower control limits for the np charts are :

$$UCL_{np} = n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})}$$

$$\text{Centre Line, } CL_{np} = n\bar{p} = \frac{\text{Total number of defectives}}{\text{Number of samples}}$$

$$LCL_{np} = n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})}$$

And draw the control limits line (CL, UCL, and LCL). Take sample numbers in the x-axis and np values in the y-axis and plot the sample and lines and points. If all the sample points lie within UCL&LCL then the process is under control.

<u>p-chart</u>

Method 1 (equal sample size)

$$p = \frac{no.of\ defective\ items\ in\ the\ sample}{Sample\ size (or) Total\ no.of\ items}$$

$$\overline{p} = \frac{\sum p}{n}$$

$$Central\ line = \overline{p}$$

$$UCL = \overline{p} + 3\sqrt{\frac{\overline{p}\left(1-\overline{p}\right)}{n}}$$

$$LCL = \overline{p} - 3\sqrt{\frac{\overline{p}\left(1-\overline{p}\right)}{n}}$$

Method 2 (un-equal sample size)

$$p = \frac{no.of\ defective\ items\ in\ the\ sample}{Sample\ size (or) Total\ no.of\ items}$$

$$\overline{p} = \frac{\sum p}{n} \quad \overline{n} = \frac{Total\ no.of\ insected}{Total\ no.of\ samples}$$

$$Central\ line = \overline{p}$$

$$UCL = \overline{p} + 3\sqrt{\frac{\overline{p}\left(1-\overline{p}\right)}{\overline{n}}}$$

$$LCL = \overline{p} - 3\sqrt{\frac{\overline{p}\left(1-\overline{p}\right)}{\overline{n}}}$$

Note:

If the lower control limit is –ve we consider the X-axis as our LCL (as our value should be only in the I quadrant).

<u>Problems:</u>

1. The following are the figures for the number of defectives of 10 samples each containing 100 items: 8, 10, 9, 8, 10, 11, 7, 9,6,12. Draw control chart for **fraction defective** and comment on the state of control of the process.

<u>Solution:</u>

Given the size of all samples are equal.

$$\text{P for sample} = \frac{No.of\ defectives\ in\ the\ samples}{No.of\ \ items\ in\ the\ samples} = \frac{8}{100} = 0.08$$

Similarly calculate p for each sample and tabulate. Divide the number of defectives by 100 to get the fraction defective.

Sample No	1	2	3	4	5	6	7	8	9	10
No.of defective	8	10	9	8	10	11	7	9	6	12
p-fraction defective	0.08	0.1	0.09	0.08	0.1	0.11	0.07	0.09	0.06	0.12

$$\overline{P} = \frac{\sum P}{n} = \frac{0.08+0.1+0.09+0.08+0.1+0.11+0.07+0.09+0.06+0.12}{10} = 0.09$$

$$UCL = \overline{P} + 3\sqrt{\frac{\overline{P}\left(1-\overline{P}\right)}{n}}$$

$$= 0.09 + 3\sqrt{\frac{0.09(1-0.09)}{100}}$$

$$= 0.09 + 3\sqrt{\frac{0.09 \times 0.91}{100}}$$

$$= 0.09 + 3(0.029)$$

$$UCL = 0.177$$

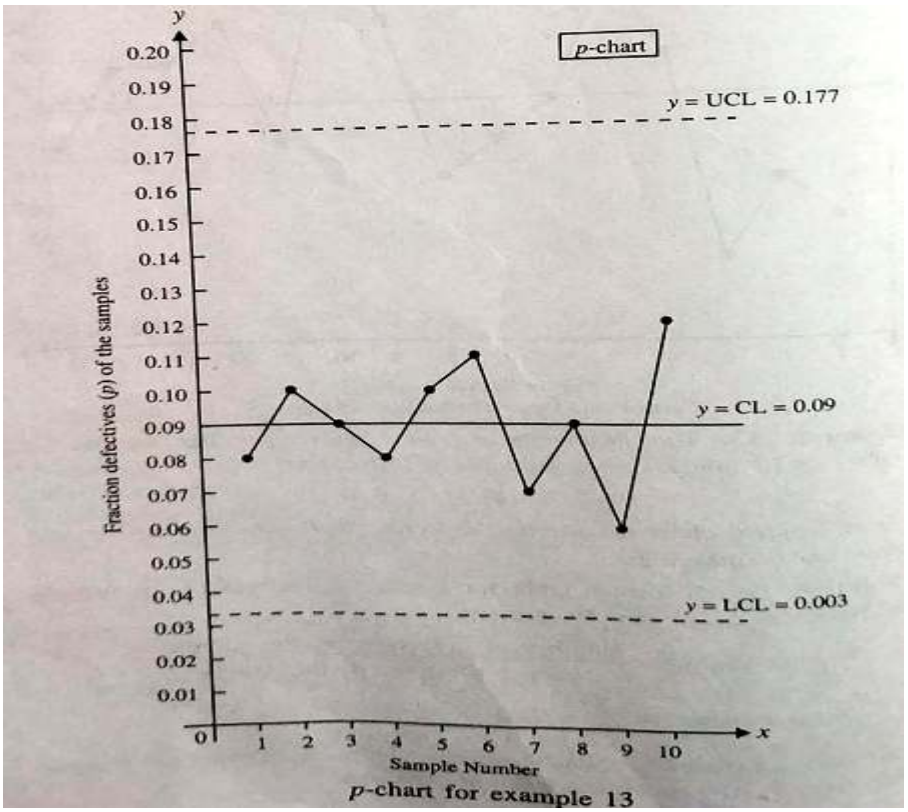
$$LCL = \overline{P} - 3\sqrt{\frac{\overline{P}(1-\overline{P})}{n}}$$

$$= 0.09 - 3\sqrt{\frac{0.09(1-0.09)}{100}}$$

$$= 0.09 - 3\sqrt{\frac{0.09 \times 0.91}{100}}$$

$$= 0.09 - 3(0.029)$$

$$LCL = 0.003$$



p-chart for example 13

Conclusion:

All values of P $are > 0.03$ and < 0.177.

i.e., All the points lies within the LCL&UCL

The process is under good control.

2. The data given below are the number of defectives in 10 samples of 100 items each. Construct a P-chart and an np-chart and comment on the results.

Sample .No	1	2	3	4	5	6	7	8	9	10
No.of defectives	6	16	7	3	8	12	7	11	11	4

Solution:

Sample size is constant for all samples, $n = 100$

P for sample $= \dfrac{No.of\ defectives\ in\ the\ samples}{No.of\ \ items\ in\ the\ samples} = \dfrac{6}{100} = 0.06$

Similarly calculate p for each sample and tabulate. Divide the number of defectives by 100 to get the fraction defective.

Sample .No	1	2	3	4	5	6	7	8	9	10
No.of defectives	6	16	7	3	8	12	7	11	11	4
P-fraction defectives	0.06	0.16	0.07	0.03	0.08	0.12	0.07	0.11	0.11	0.04

Total number of defectives= $6+16+7+3+8+12+7+11+11+4 = 85$

Total number of inspected $= 10 \times 100 = 1000$

$$\overline{P} = \frac{85}{1000} = 0.085$$

For P-chart

$$UCL = \overline{P} + 3\sqrt{\frac{\overline{P}\left(1-\overline{P}\right)}{n}}$$

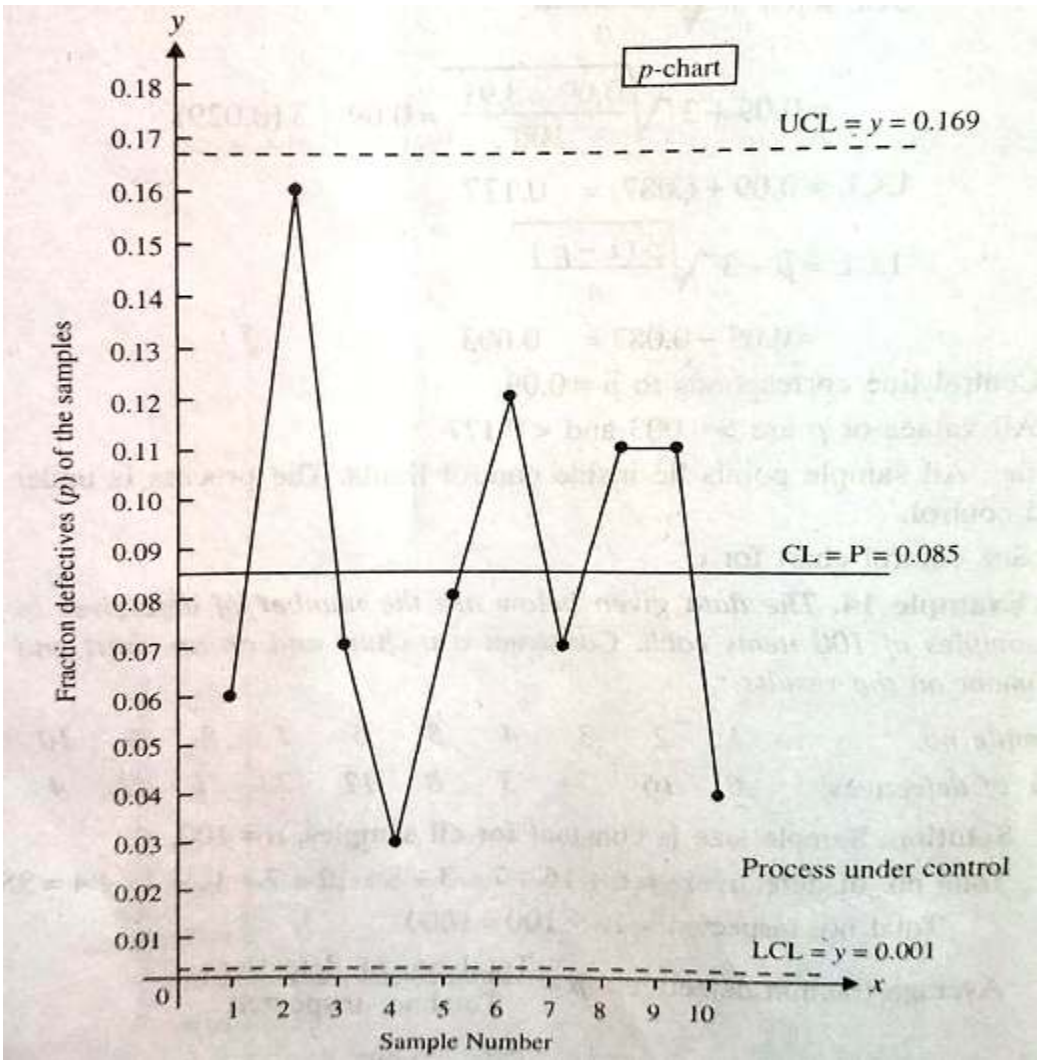
$$= 0.085 + 3\sqrt{\frac{0.085\left(1-0.085\right)}{100}}$$

$$= 0.085 + 3\sqrt{\frac{0.085 \times 0.915}{100}} = 0.1687$$

$$LCL = \overline{P} - 3\sqrt{\frac{\overline{P}(1-\overline{P})}{n}}$$

$$= 0.085 - 3\sqrt{\frac{0.085(1-0.085)}{100}}$$

$$= 0.085 - 3\sqrt{\frac{0.085 \times 0.915}{100}} = 0.0013$$



Conclusion :

All the values are less than UCL=0.1687 and greater than LCL=0.0013.

In the control chart, all sample points lie within the control limits.

The process is control.

<u>For np-chart</u>

$$UCL = n\overline{P} + 3\sqrt{n\overline{p}(1-\overline{p})}$$

$$= 100 \times 0.085 + 3\sqrt{100 \times 0.085(1-0.085)}$$

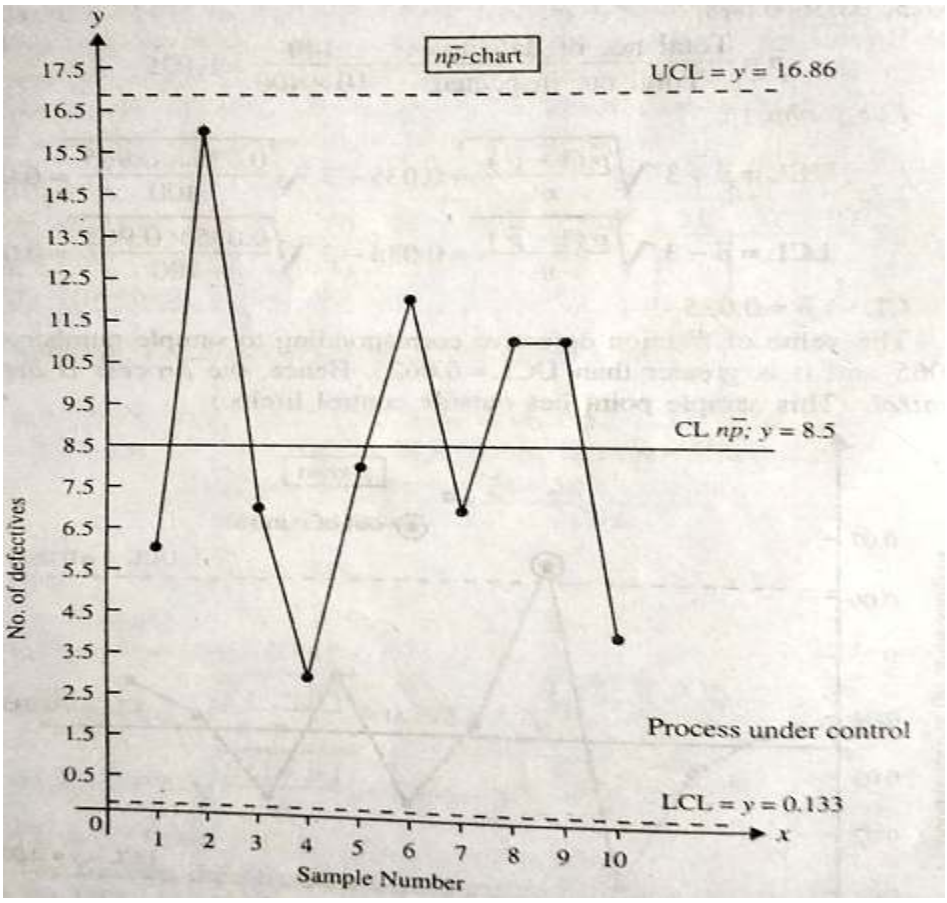
$$UCL = 16.87$$

$$LCL = n\overline{P} - 3\sqrt{n\overline{p}(1-\overline{p})}$$

$$= 100 \times 0.085 - 3\sqrt{100 \times 0.085(1-0.085)}$$

$$LCL = 0.13$$

$$n\overline{P} = 100 \times 0.085 = 8.5$$



Conclusion

All the values of number of defectives in the table lie between UCL and LCL.

The process is under control.

3. Inspection of 10 samples of size 400 each from 10 lots of bolts revealed the following number of defective units.

Sample No.	1	2	3	4	5	6	7	8	9	10
No.of defectives	3	2	0	4	3	8	10	4	4	5

Check if the process is under control or not using p-chart.

Solution:

$$p = \frac{no.of\ defective\ items\ in\ the\ sample}{Sample\ size(or)Total\ no.of\ items} = \frac{3}{4000} = 0.00075$$

Sample No.	1	2	3	4	5	6	7	8	9	10
No. of defectives	3	2	0	4	3	8	10	4	4	5
P	0.00075	0.0005	0	0.001	0.00075	0.002	0.0025	0.001	0.001	0.00125

$$\overline{p} = \frac{\sum p}{n} = 0.001075$$

$$UCL = \overline{p} + 3\sqrt{\frac{\overline{p}(1-\overline{p})}{n}}$$

$$= 0.001075 + 3\sqrt{\frac{0.001075(1-0.001075)}{4000}}$$

$$UCL = 0.001075 + 3(0.000518) = 0.002629$$

$$LCL = \overline{p} - 3\sqrt{\frac{\overline{p}(1-\overline{p})}{n}}$$

$$= 0.001075 - 3\sqrt{\frac{0.001075(1-0.001075)}{4000}}$$

$$= -0.00047$$

Conclusion

Take x-axis as the LCL.The process is under control.

4. In a factory producing spark plugs, the number of defectives found in the inspection of 10 lots of 100 each is given below: Draw the control chart for the number of defectives and comment on the state of control.

Sample Number	1	2	3	4	5	6	7	8	9	10
No. of defective (np)	17	15	14	26	9	4	19	12	9	15

Solution

n=100

Total no. of inspected items= $10 \times 100 = 1000$

Total no. defective $= 17 + 15 + 14 + 26 + 9 + 4 + 19 + 12 + 9 + 15 = 140$

P value= $\frac{140}{1000}$

$\overline{p} = 0.14$

$$UCL = \overline{p} + 3\sqrt{\frac{\overline{p}(1-\overline{p})}{n}} = 0.14 + 3\sqrt{\frac{0.14(1-0.14)}{100}} = 0.14 + 3(0.0346) = 0.2438$$

$$LCL = \overline{p} - 3\sqrt{\frac{\overline{p}(1-\overline{p})}{n}} = 0.14 - 3\sqrt{\frac{0.14(1-0.14)}{100}} = 0.14 - 3(0.0346) = 0.0362$$

Using p-chart the process is out of control

Np-chart

$n\overline{p} = 100 \times 0.14 = 14$

$$UCL = n\overline{p} + 3\sqrt{n\overline{p}(1-\overline{p})} = 14 + 3\sqrt{14(1-0.14)} = 14 + 3(3.469) = 24.407$$

$$LCL = n\overline{p} - 3\sqrt{n\overline{p}(1-\overline{p})} = 14 - 3\sqrt{14(1-0.14)} = 14 - 3(3.469) = 3.593$$

$CL = n\overline{p} = 14$

One point lies outside the control limit

The process is out of control in np-chart.

DISCUSSION QUESTIONS:

1. What do you mean by central limit ?

2. Define Lower and upper control limits

3. When you say the limit is out of control ?